Loss-Free Balancing + Sigmoid Gating

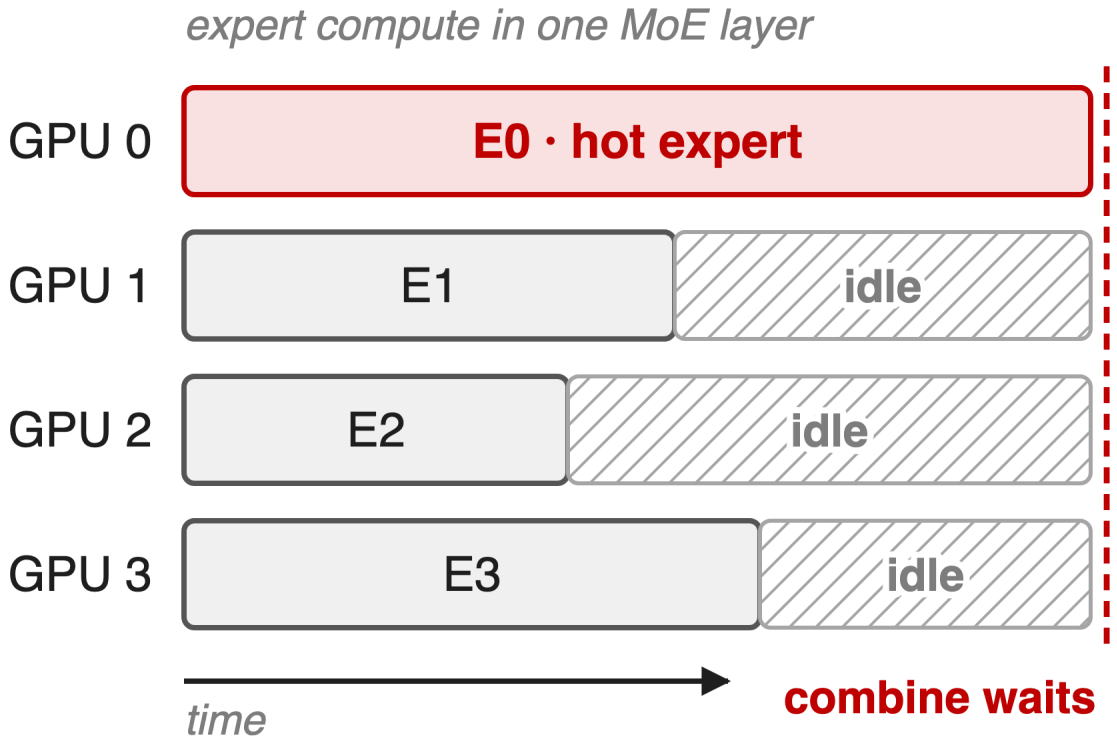
Balance MoE experts with a routing bias, not a loss

DeepSeek-AI (arXiv 2408.15664) · GLM-4.5 · DeepSeek-V3/V4 · Megatron-Core

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Every MoE step waits for its busiest expert

- DeepSeek-V3: **256** routed experts, top-**8**, spread over **64** GPUs
- GLM-4.5: **160** routed experts, top-**8**
- Starved experts stop learning: routing collapse



A tuned aux loss still leaves 72% overload

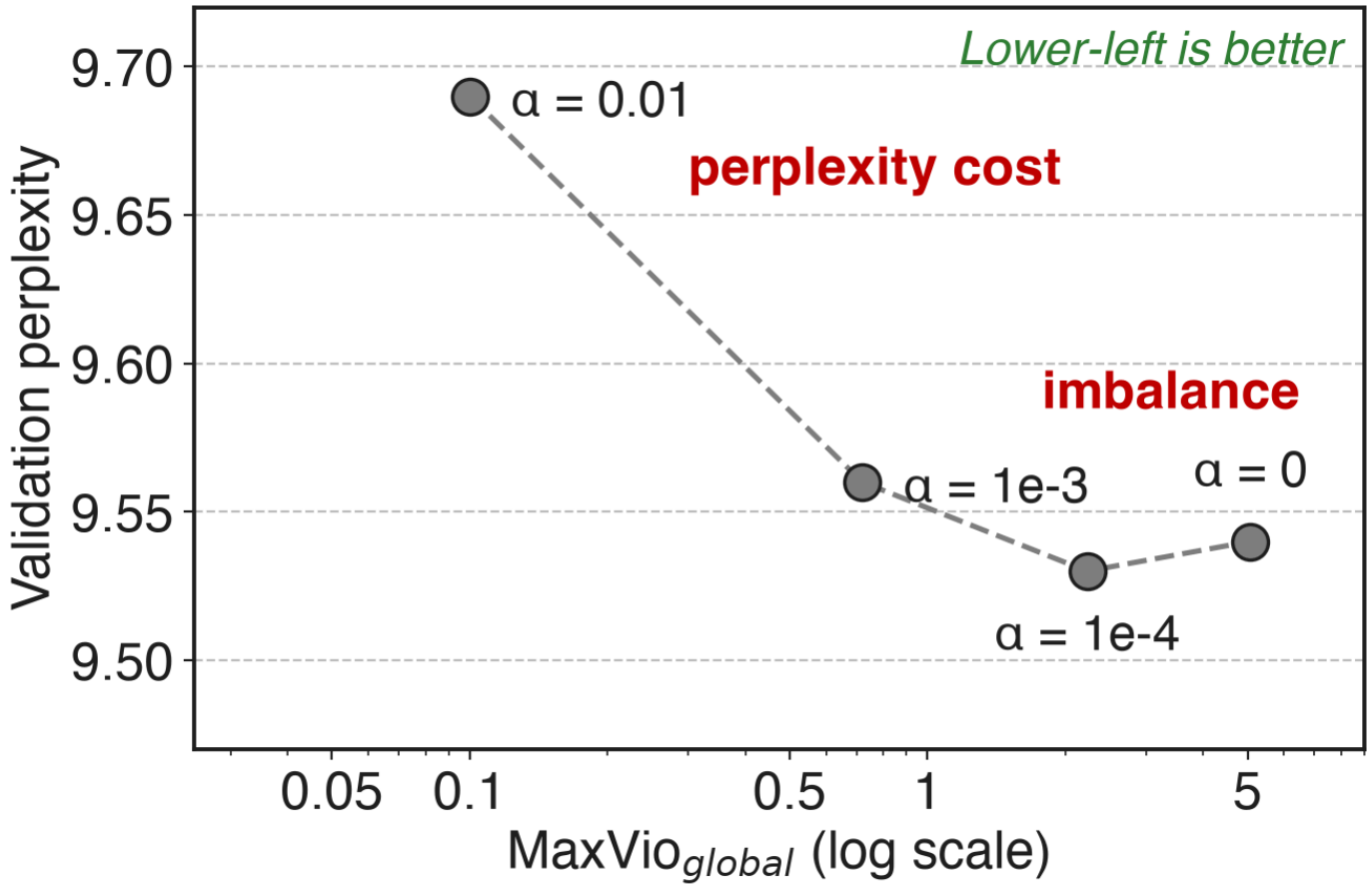
+72%

more tokens than its fair share on the busiest expert, even with a tuned auxiliary loss (1B MoE, $\alpha = 0.001$)

arXiv 2408.15664, Table 2: MaxVio_global = 0.72, i.e. max load = 1.72 × mean load (layer average)

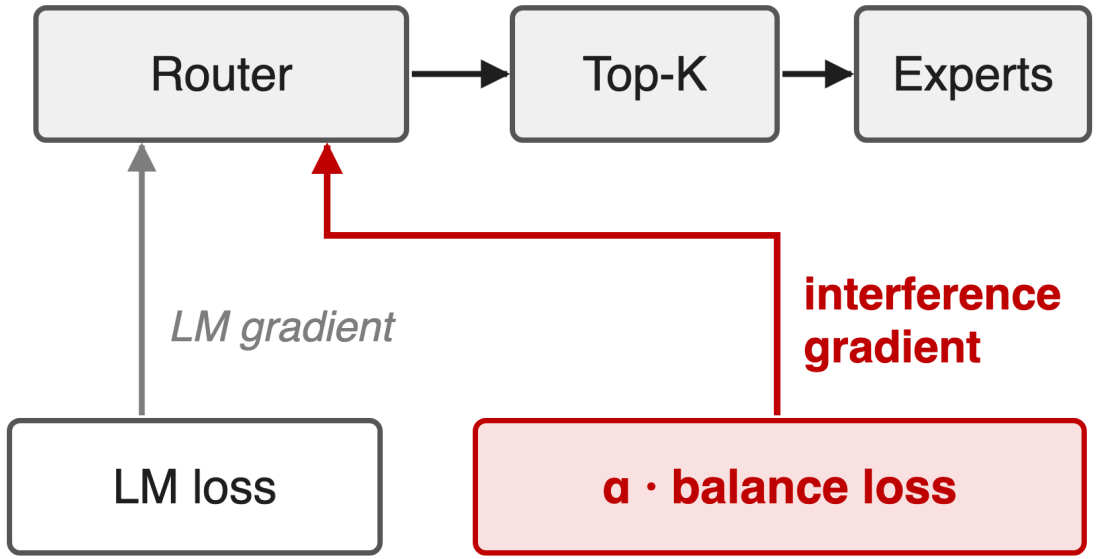
Every aux-loss weight loses balance or perplexity

- **α = 0**: MaxVio ≈ **5**, near collapse
- **α = 0.01**: balanced, perplexity ≈ **9.69**
- No α reaches the lower-left corner

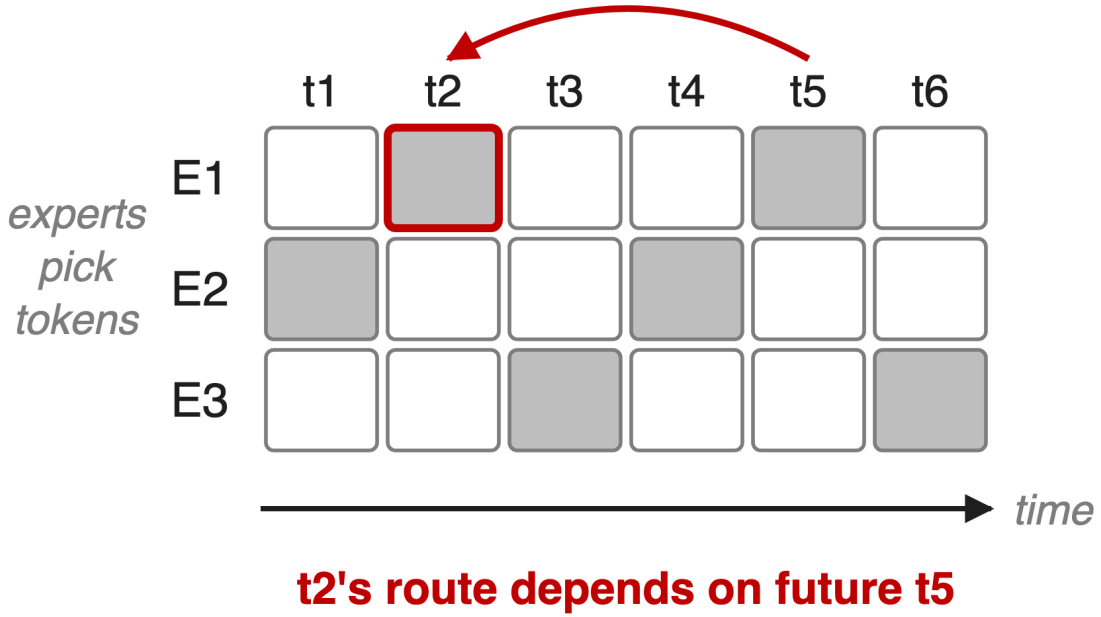


1B MoE. α = 1e-3 from Table 2; other α read off Fig. 2 of arXiv 2408.15664 (approximate)

Existing fixes touch the gradient or the future



(a) Auxiliary balance loss



(b) Expert Choice routing

Both couple balance to the gradient or to future tokens

Insight: top-K selection needs no gradient

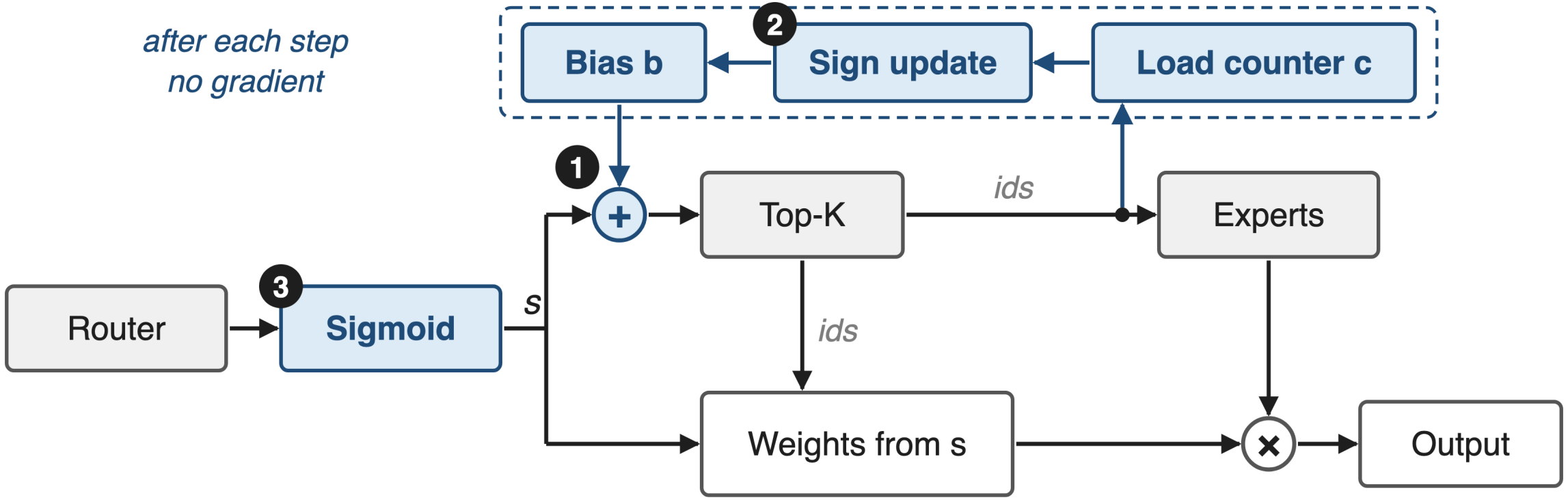
- Router scores do two jobs: **pick** experts, **weight** their outputs
- Balance needs only the pick, and top-K is never differentiated

Insight: top-K selection needs no gradient

- Router scores do two jobs: **pick** experts, **weight** their outputs
- Balance needs only the pick, and top-K is never differentiated

Steer the pick with a per-expert bias that never enters the loss or the output

Bias the pick, keep the weights, close the loop



❶ biased Top-K · ❷ sign update after each step · ❸ sigmoid scores; bias and counters carry no gradient

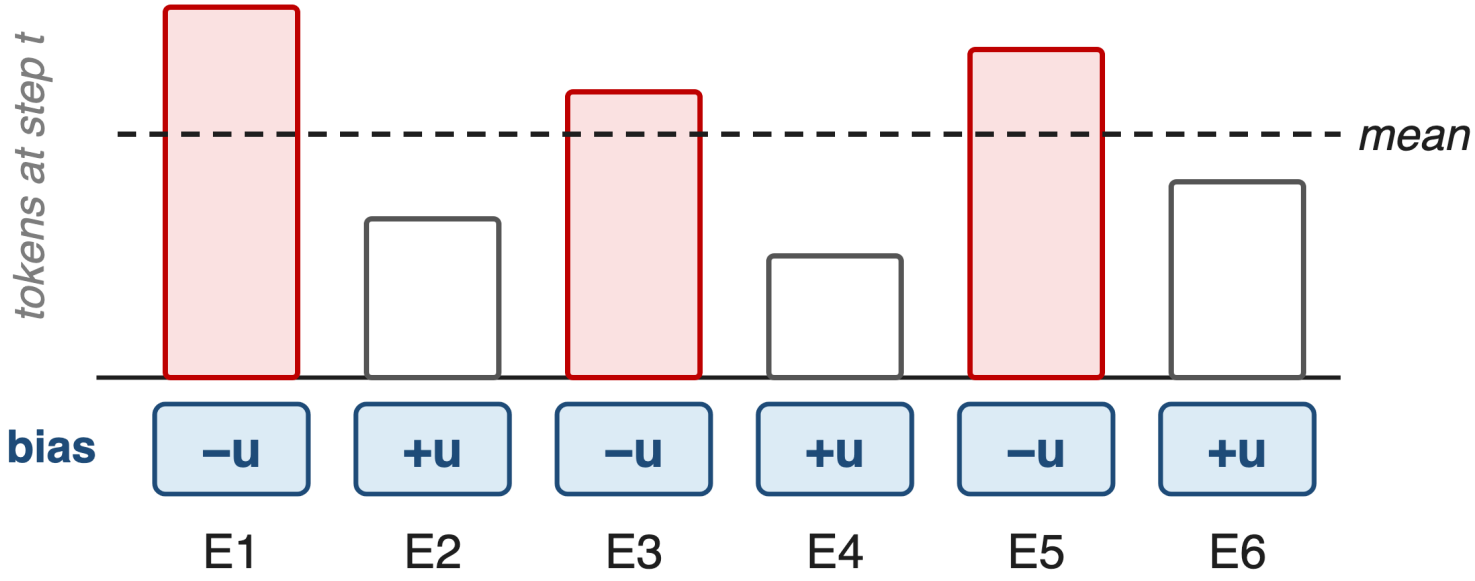
❶ The bias moves the pick, never the weight

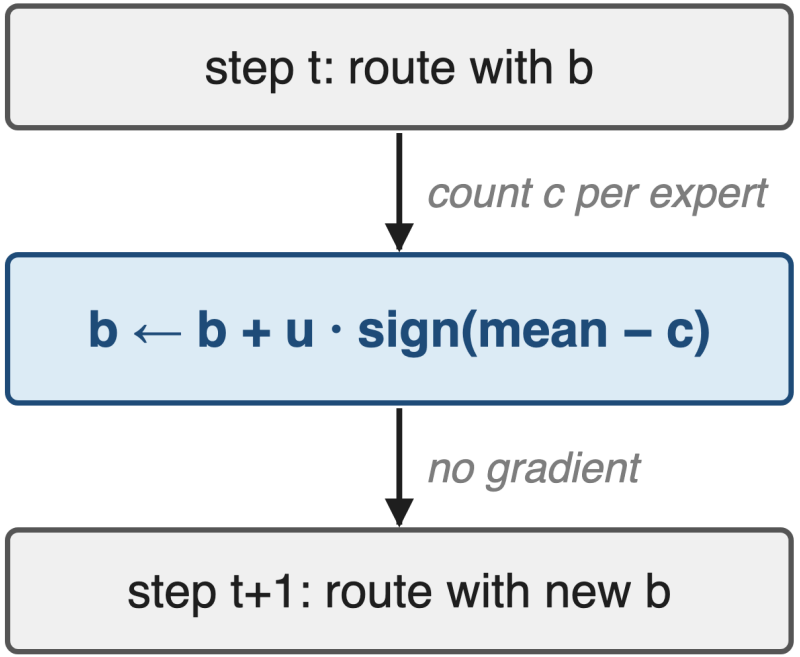
	E1 · hot	E2	E3	E4
score s	0.90	0.85	0.60	0.20
bias b	–0.30	0	+0.10	0
s + b	0.60	**0.85**	**0.70**	0.20
top-2 pick	–	**selected**	**selected**	–
gate weight	–	**0.59**	**0.41**	–

weights use s only

Without b, top-2 is E1 + E2; with b, E2 + E3, weighted by s alone

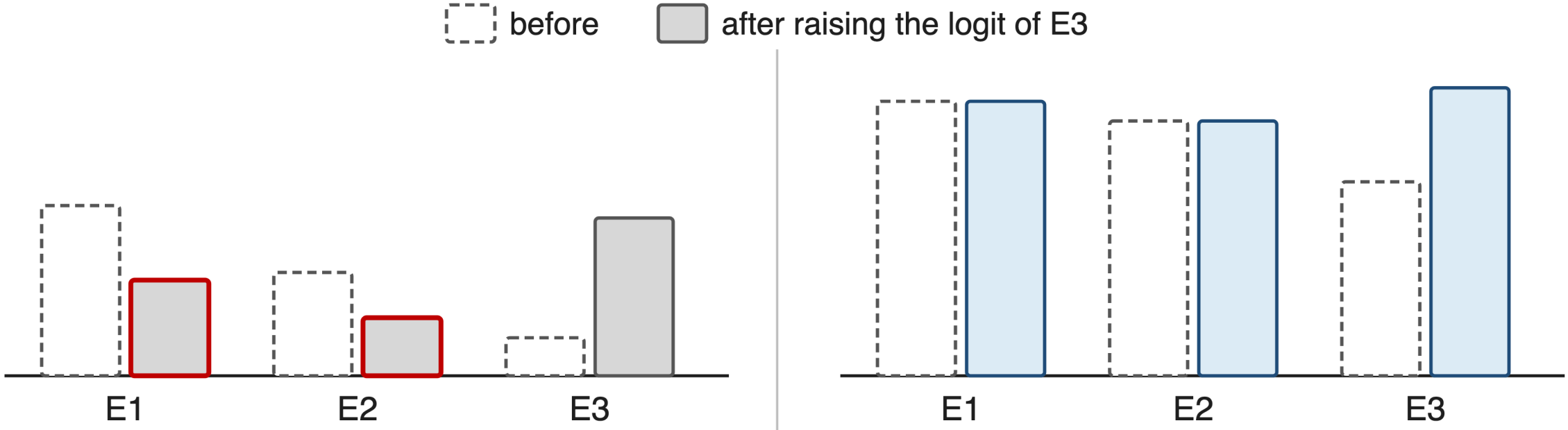
❷ A sign step nudges each bias by ±u





u = 0.001: smaller adapts too slowly early, larger oscillates late

❸ Sigmoid keeps each expert's score independent



(a) Softmax: E1 and E2 shrink

(b) Sigmoid: E1 and E2 unchanged

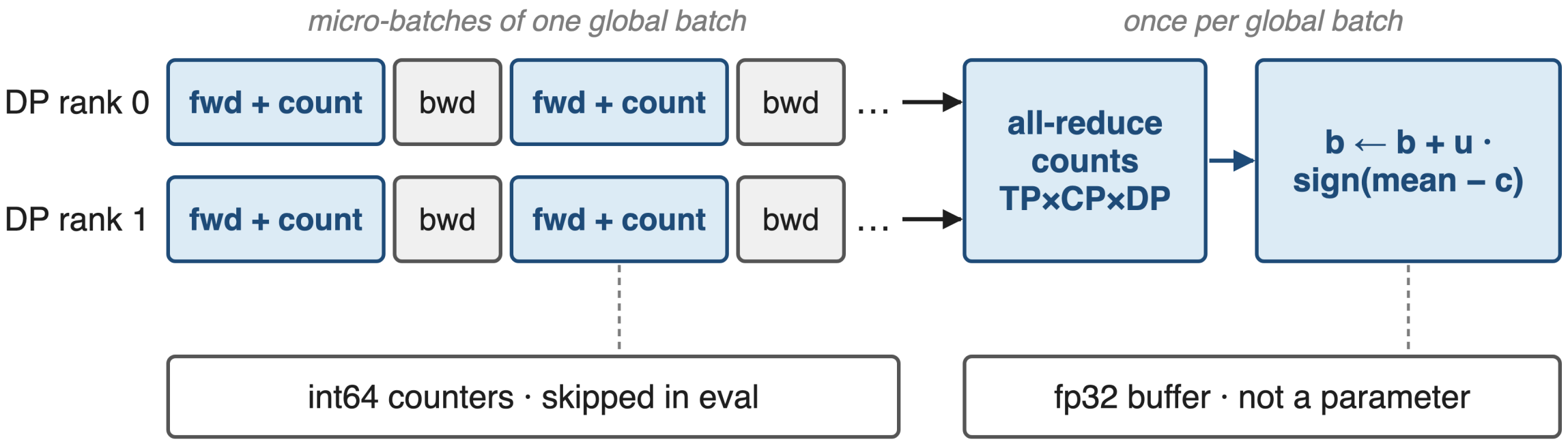
Softmax couples the score gaps a fixed bias must shift; sigmoid does not

Production: u = 0.001 plus a 1e-4 sequence loss

	DeepSeek-V3	GLM-4.5	DeepSeek-V4
Score function	Sigmoid	Sigmoid	√Softplus
Routed experts	256 · top-8	160 · top-8	256–384 · top-6
Bias update rate	**0.001**	**0.001**	**0.001**
Rate set to 0	last 500B tokens	after 15T tokens	*not stated*
Sequence loss weight	**1e-4**	**1e-4**	**1e-4**

All three keep a tiny sequence loss against in-sequence skew

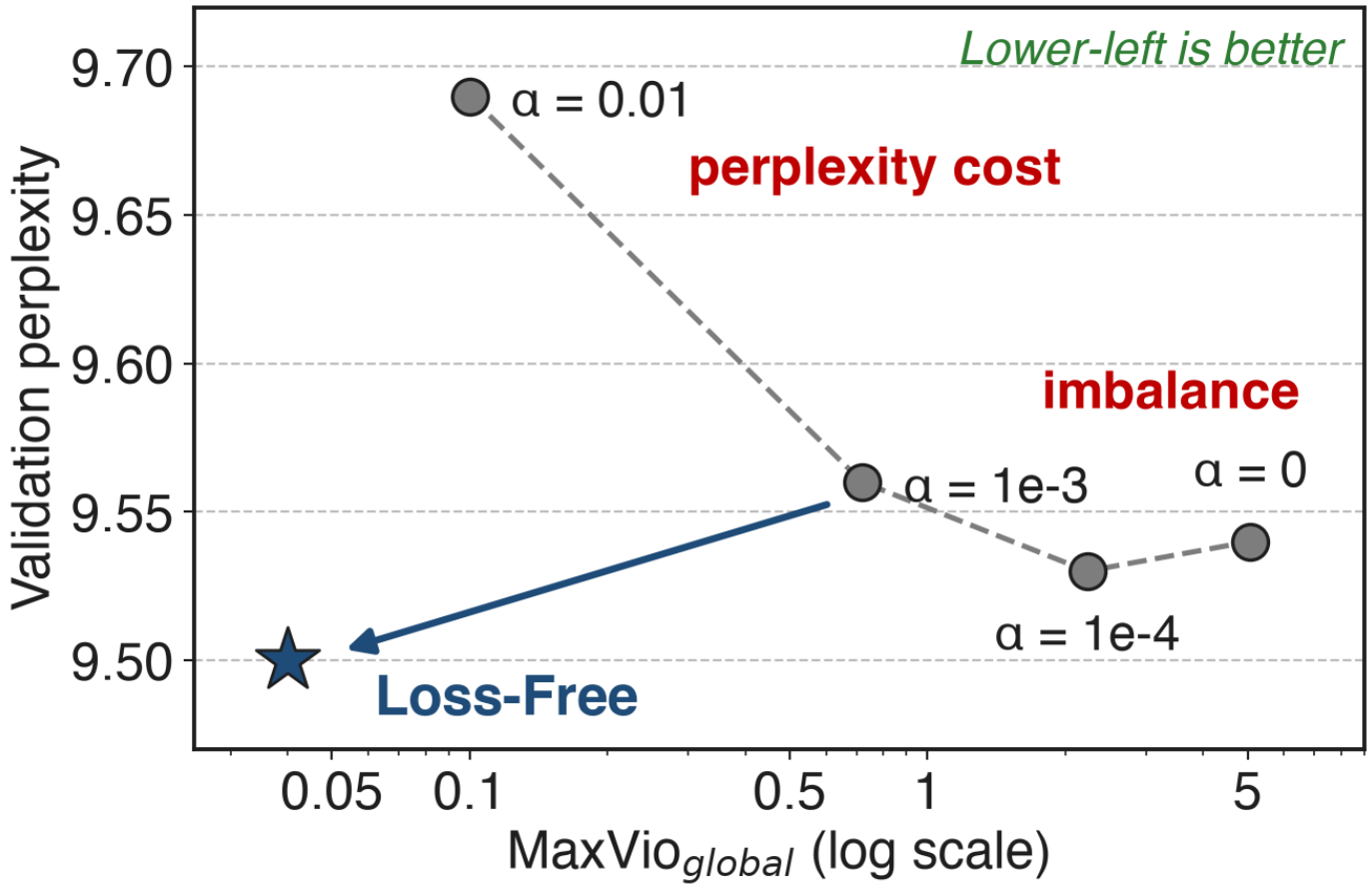
MCore updates the bias once per global batch



--moe-router-score-function sigmoid --moe-router-enable-expert-bias --moe-router-bias-update-rate 1e-3 (fused router: deck 42)

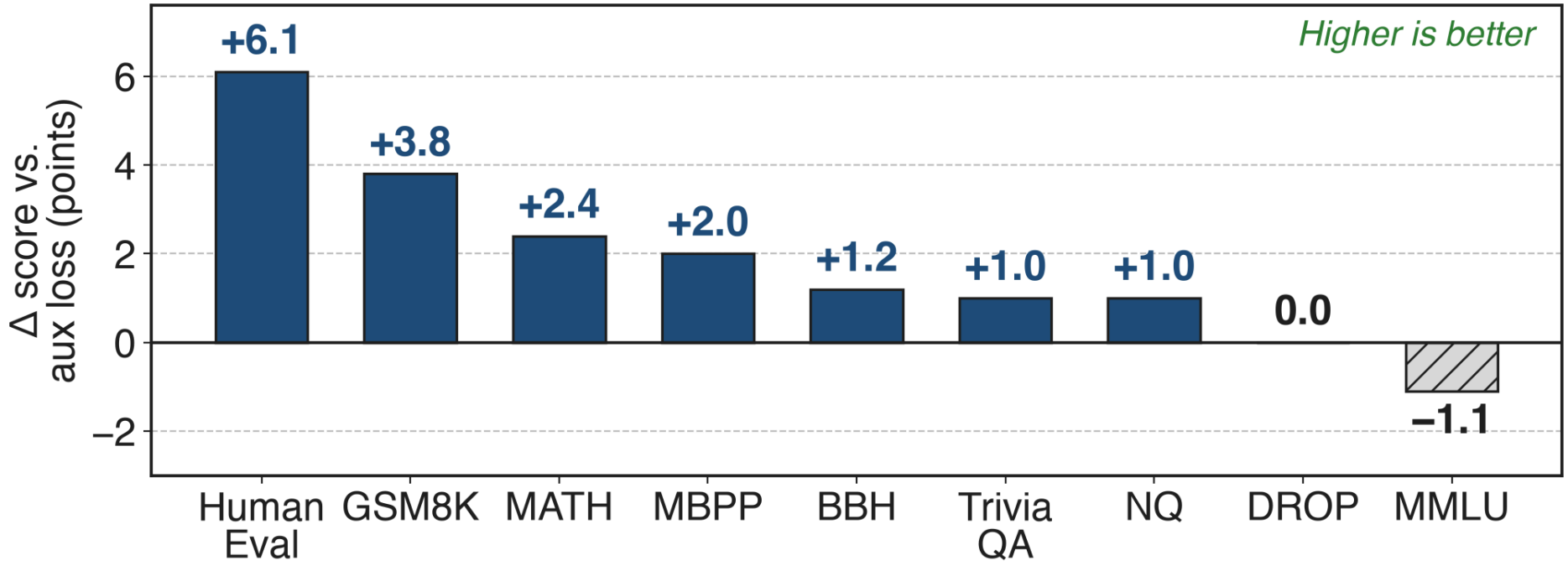
MaxVio 0.72 → 0.04, with lower perplexity

- 1B: perplexity **9.56 → 9.50**
- 3B: **7.97 → 7.92**, MaxVio **0.52 → 0.04**
- Balance keeps improving as EP batches grow



1B on 100B tokens, 3B on 200B tokens; α = 1e-2, 1e-4, 0 read off Fig. 2 (approximate)

At 228.7B, loss-free wins 7 of 9 benchmarks



DeepSeek-V3 ablation, 228.7B total / 20.9B active MoE on 578B tokens, same data; Pile BPB 0.656 → 0.652

Up to +6.1 HumanEval and +3.8 GSM8K; MMLU drops 1.1

Limits: batch scope, one step size, ~1k experts

- **Per batch, not per sequence**: production adds the **1e-4** loss
- Scope explains part of the gain: a batch-wise aux loss ties (**2.253**)
- **One fixed step u**: slow or oscillating
- **896** experts per layer: Kimi K3 replaces the sign rule (deck 56)

Summary

- Aux loss trades balance for perplexity: **+72%** overload at the tuned α
- **Insight**: top-K needs no gradient, so steer it with a bias
- Sign step on token counts; weights still from sigmoid scores
- MaxVio **0.72** → **0.04**, perplexity **9.56** → **9.50**; production u = **0.001**

https://arxiv.org/abs/2508.06471